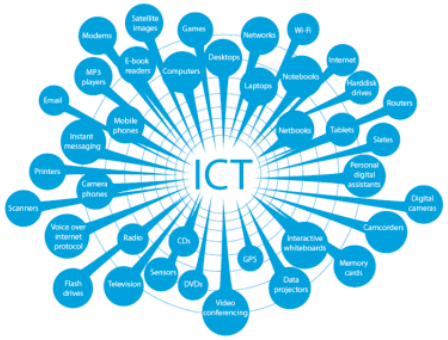




KARATINA UNIVERSITY
EMERGING TRENDS IN ICT
BIT 123



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Preamble

- ICT is always improving and changing and new technologies are being developed all of the time.
- Developments in technology will, by nature, impact on our everyday lives and this section discusses some of these.

- The Information and Communications Technology (ICT) industry is experiencing profound change across a broad range of areas. The following evolving trends play a significant role and impact on society and therefore ICT:

Catalysts....

- **Anytime, anywhere, any device access to communications, content, commerce, and applications:** Consumer Cloud Services is a concept involving the storage of various consumer electronic items in “the cloud” for the benefit of anytime, anywhere, any device access.
- The concept is that one would not be limited to things like not being able to listen to their music because it is stored on a laptop (only) and not accessible when wanted on the go.

Catalysts….

- **Increased emphasis on non-human communications:** With worldwide carriers reaching saturation of human users, network operators are looking for new ways to generate revenue and enhance profitability. At the same time, miniaturization is reaching the point in which embedded computing can be virtually ubiquitous. With the advent of IP version six (IPv6) and the attendant huge increase in IP addresses, virtually everything can be labeled (e.g. have an address or reference). All of this adds up to translate into a world in which objects (devices, equipment, and other assets) are communicating, transacting, signaling, etc.
- Arguably this market will overtake human communications rapidly as there are vastly more objects in the world than humans, they can communicate much faster, much more frequently, and for many more reasons than humans.

Catalysts….

- **The convergence and integration of many things:** Many things are converging including everything from networks, services, and entire business models. One key example that is particularly poignant for enterprise is the convergence of M2M, IoT, Social, and Big Data. In many enterprises, discrete information systems that were developed over time for specific functions in an uncoordinated fashion have traditionally hampered high-level decision-making. When internal systems are organized in a silo fashion, top management typically struggles to spot correlations among individual departmental reports that could indicate areas for cross-departmental or divisional improvements. Historically, discrete data collection systems fail to present big picture perspectives since any actionable information on trends is constrained, as if perceived through horse blinders.

Catalysts….

- **Open networks, open interfaces, and many applications:** While a large part of the market is yet to understand their technological and strategic significance, Telco Network Application Programmer Interface (API) play a critical part in carrier networks as a secondary stream of revenue. Telco Network APIs capitalize on existing network infrastructure to create a vast array of business opportunities for carriers worldwide.
- In essence, these APIs allow carriers to disseminate a wealth of internal information or resources to third parties. This could entail everything from network QoS for video service delivery to Subscriber Data Management (SDM) for advertising and profiling, the goal being to enable third party developers to offer services in return for revenues.

Emerging Technologies/Trends

Definition: ***"Emerging/New technologies are those that are currently being developed or will be developed in the next 5 to 10 years, and which will alter the business and social environment."***

What we shall cover

- The technologies (and their impacts on everyday life) that we are going to focus on:

1. **Artificial intelligence**
2. **Biometrics and RFIDs**
3. **Robotics**
4. **Internet of Things**
5. **Quantum cryptography**
6. **Computer assisted translation and Computer Vision**
7. **3D and holographic imaging**
8. **Affectiva**
9. **Cloud computing**
10. **Virtual reality**
11. **Haptics**
12. **Ambient Intelligence**
13. **Big data**

Artificial **I**ntelligence (AI)

- focused on creating computer systems that **simulate human intelligence**.
- The term was first used in 1956 by a computer scientist at the Massachusetts Institute of Technology (MIT) who was focused on trying to **make computers behave like humans**.

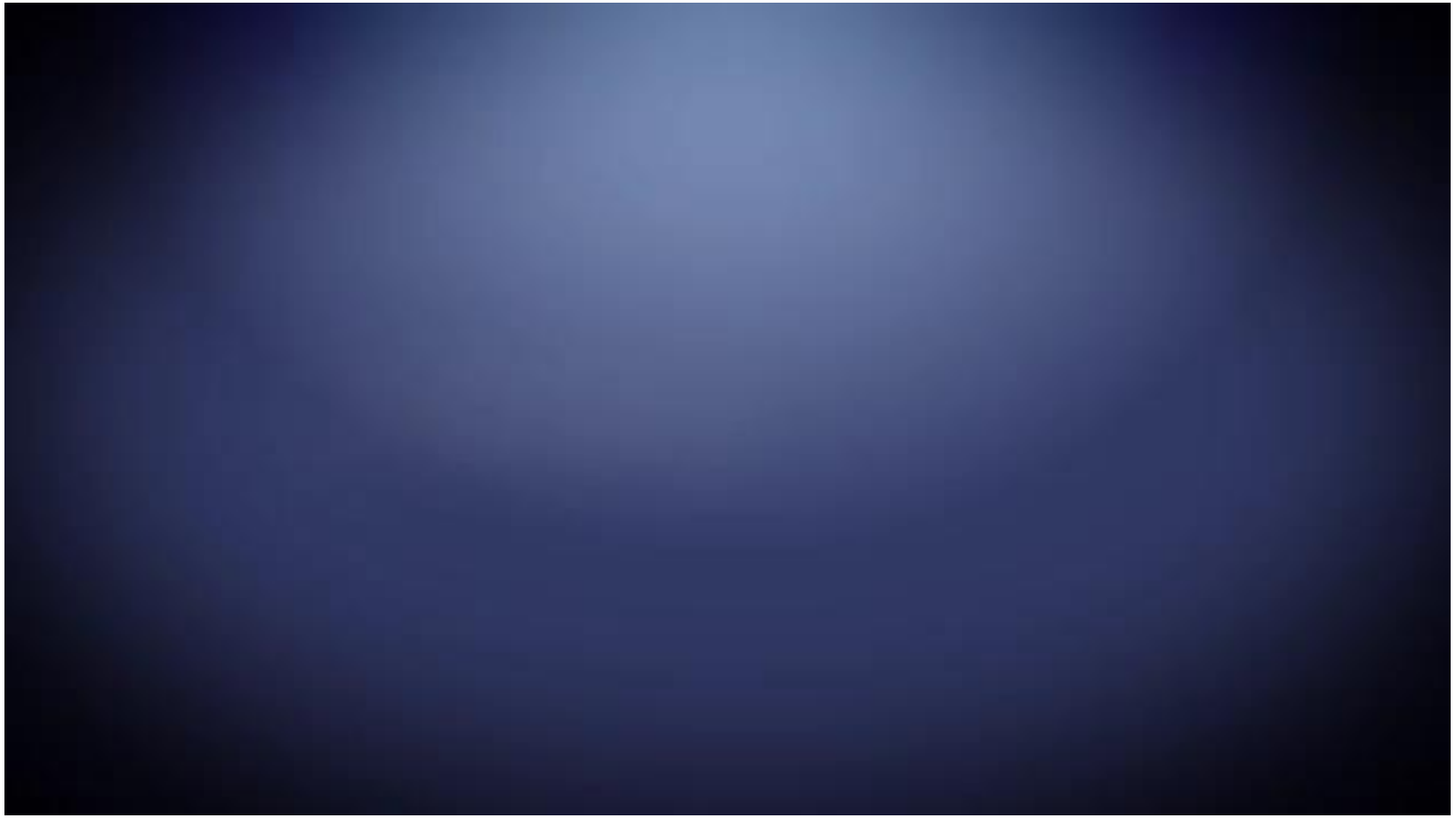
AI is being developed in the following areas:

Game Playing -	Computers developed to **play games against human players**. **For example:** In 1997 a computer named **'Deep-Blue'** defeated a world champion in the **game of chess**.
Expert Systems -	These are computers that have been **programmed to make decisions** based on information they are given. **For example:** Medical expert systems can **diagnose patient's illnesses** based on symptoms entered.
Languages -	This type of AI involves computers that can **understand different human languages** as they are spoken to them.
Robotics -	Robotic artificial intelligence is where machines are programmed to **imitate a human**.

Impacts of AI on everyday life:

Accurate prediction of weather	AI software will soon be used to sift through weather data more accurately that humans can and will be used to **predict approaching storms** and **automatically issue warnings**.
Increased leisure time	Robotic vacuum cleaners are becoming more and more popular. These can **detect walls** and other objects in order to **vacuum around them**. People can **leave them running** whilst they enjoy extra spare time.
Safer transport	Cars that can **self-park** already exist and it is predicted that **cars that drive themselves** will be available soon. These could drastically **reduce road accidents**. Driverless trains already exist in some countries!
Increased Personal safety	Modern home alarm systems use artificial intelligence software that can tell the difference between the **home owners and intruders**. The software **automatically alerts the police** when intruders are detected.
Improved medical care	**Robotic surgery assistants** are being used to **quickly** and **accurately** pass the correct surgical tools to doctors. The **few seconds saved** in getting the correct tool to the doctor can save patient's lives.

Robo surgeon



Driverless Train



Biometrics

- Biometrics is where **parts of a person's body** are used for **identification purposes**. Examples are shown in the table below:

Fingerprints -	These are **impressions** embedded at the **end of human fingers** and **thumbs**. Fingerprints kept in a database can be matched to those left at crime-scenes to help identify the culprit.
Eye recognition -	Eye scans **analyse the iris** which is the coloured ring that surrounds the pupil.
Face recognition -	This is where the **shapes of individual's faces** are analysed.
Voice recognition -	**Pitch, tone** and **frequency** of voices are unique and can be analysed to identify people.

Biometrics

- All of these parts of the human body are **unique from person to person** and can be used to **authenticate identity**.

Note: Even **identical twins** have slightly different fingerprints and voices etc.

- Before biometric methods can be useful, people have to perform a process known as **'biometric enrolment'**.

This is where body-part data such as fingerprints or voice patterns are **captured and stored** within the system so that they can be used to identify the person later on.

- Biometrics are beginning to be used **in place of passwords and physical locks** as a means of **security**.
  Biometrics have advantages over these older methods as **body parts cannot be lost**, **forgotten** or **stolen** as can be the case with passwords and keys.

Impacts of Biometrics on everyday life

Better airport security	Iris recognition is already in use in some **airports**. Travelers have their eyes and iris scanned into a system and this data is **later matched up** when the person is performing **airport checks**.
Increased building security	**Fingerprint access to buildings** have been replacing the older methods of locks and keys. This methods ensures that **only authorised people** can enter restricted buildings or rooms.
Reduced car theft	Cars already exist that use fingerprints to only **unlock their doors** or **start the engine** for the **fingerprint** that is **registered**. This means that the doors will not unlock for a print that is not recognised and makes the car **harder to steal**.
More secure mobile phones	Mobile phones contain our lives. We used our phones for everything from social media to shopping online. They need to be as secure as possible in order to protect the valuable data that they contain. Apple recently released an **iPhone model** that uses a **fingerprint reader** to identify the **true owner** of the phone. It will not unlock for a fingerprint that it does not recognise.

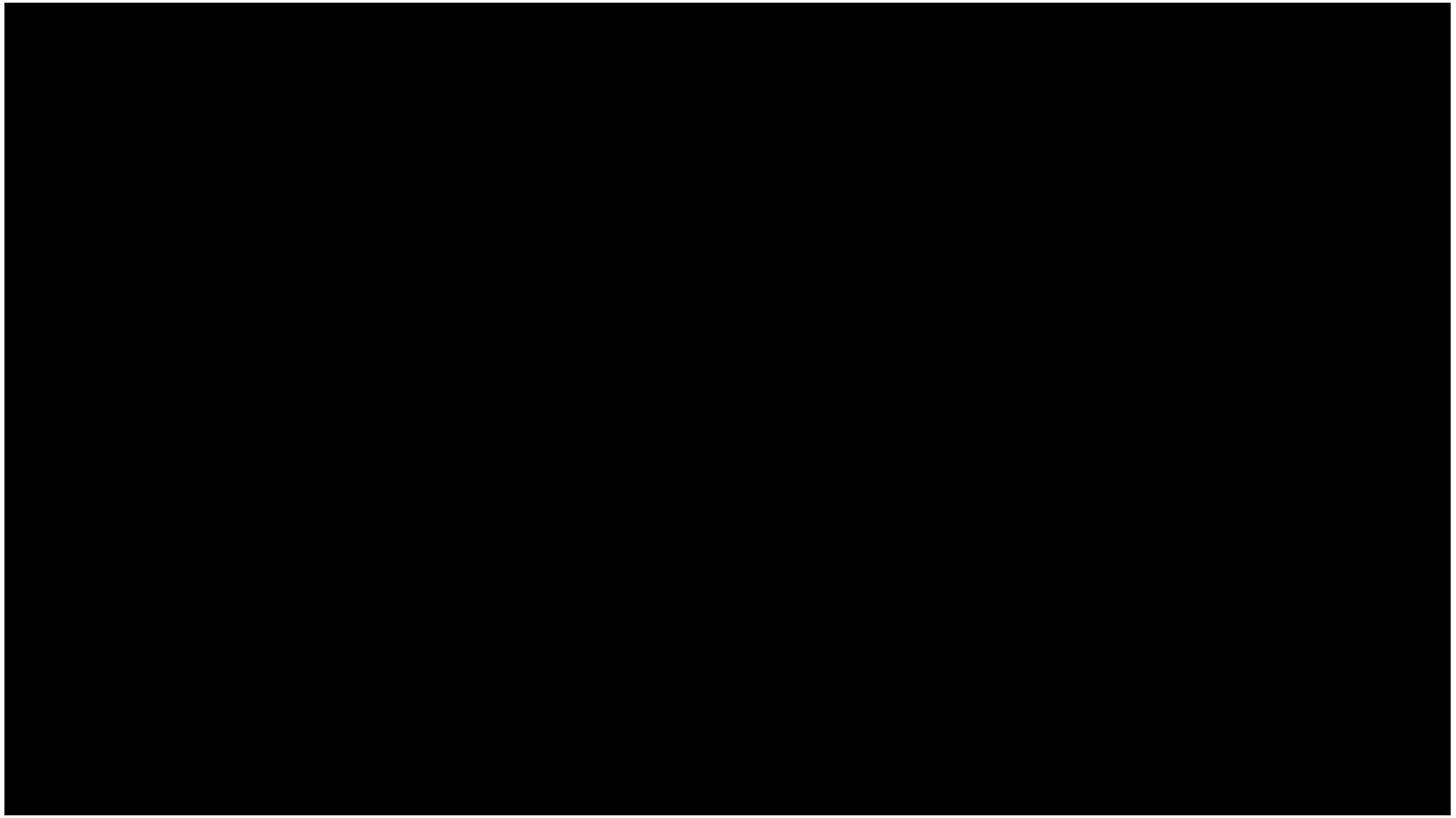
Finger Print Door lock

★ Biometric Fingerprint Door Lock "Aegis"

★ Deadbolt, Right Handed Installation



Iris Airport recognition system



Robotics

- Robots are used to perform a wide range of **physical tasks.** They are either **automated** (controlled by a computer chip) or **manually controlled** by a human.

- There are 4 different types of robots:

•**Manufacturing robots** (used to perform repetitive tasks such as welding) •**Carrier robots** (used by the military to carry heavy loads over dangerous terrain) •**Domestic robots** (used in homes to perform cleaning tasks such as vacuuming) •**Exploration robots** (used to visit and send images from places such as Mars)

Some more typical tasks that robots can be used for are;

Dangerous jobs -	E.g. **disposing of bombs**, **spray painting** or cleaning up **nuclear waste**. **Note:** these are all jobs that could harm or kill a human.
Exploring extreme environments-	E.g. inside **volcanoes**, **planets** or the **depths of the ocean**. **Note:** humans cannot visit these environments due to lack of oxygen and high pressure / heat levels.
Repetitive manufacturing jobs -	E.g. **production lines**, **packing** and **welding** etc. **Note:** these jobs can also be performed by humans but robots can do them much faster and more efficiently.
Moving heavy objects -	E.g. installing **large engines**, **moving pallets** of items etc.

Search and Rescue robots



Bomb Defusing Robot



3D and Holographic Imaging (aka holograms)

- This is a technique where images are made to **appear three-dimensional** and to actually have **depth. The** two-dimentional images need to have been **shot at different angles.**
- Two different types of **laser beams** are used to record the two-dimensional images onto a **single photographic plate**.
- This creates **one single image** that incorporates the angles of the original two-dimensional images. This produces a 3D effect.
- When viewing the image, human eyes see it from slightly different angles. The brain combines them into a three-dimensional image.

Impacts of 3D Imaging on everyday life

Improved security

Credit cards, **ID cards**, **software** and some **bank notes** include holograms as a way of trying to **prevent forged duplicates** being created.

NOTE: Forgeries don't usually include a hologram as they are **difficult** and **expensive to reproduce**.

Better movie experiences

Hollywood have been using 3D imaging within the **production of movies** for many years.

These provide the viewer with a much more **immersive experience**.

NOTE: 3D movies require the viewer to wear **special glasses** for the effect to take place. The glasses project two images shot at different angles (one in each eye) and your brain puts them together as one 3D image.

Greater data storage

It is thought that the technology behind holograms will eventually be used to provide the means to **store large amounts of data**.

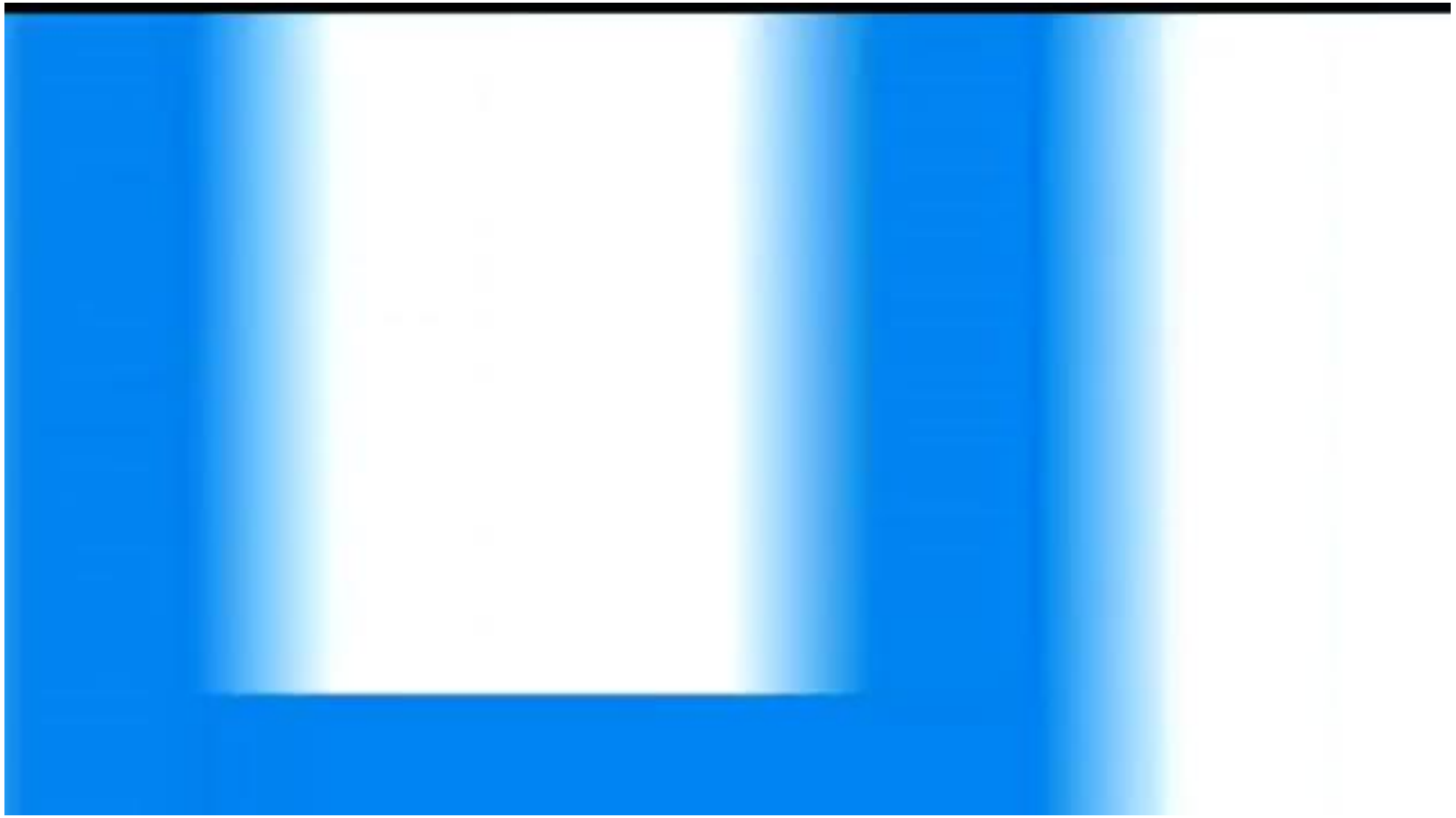
Companies have already produced discs that use holographic layers that each have the potential to hold a massive **3.9 terabytes**.

NOTE: This is the equivalent of over **150 standard Blu-ray discs**.

Holograms in security



Holograms in bank notes



Virtual Reality

- Virtual reality is where computers are used to **create an artificial environment** that users can **interact with** as if it were real.
- Virtual reality is not really meant for gaming purposes. It is used for more **serious purposes** such as:
  - **Allowing architects to walk around a virtual version of their design** (this gives a better idea of what the finished building will look like)
  - **Training soldiers in combat** (flight simulation, battlefield simulation)
  - **Training surgeons** (virtual patients can be operated on to provide experience to trainee surgeons)

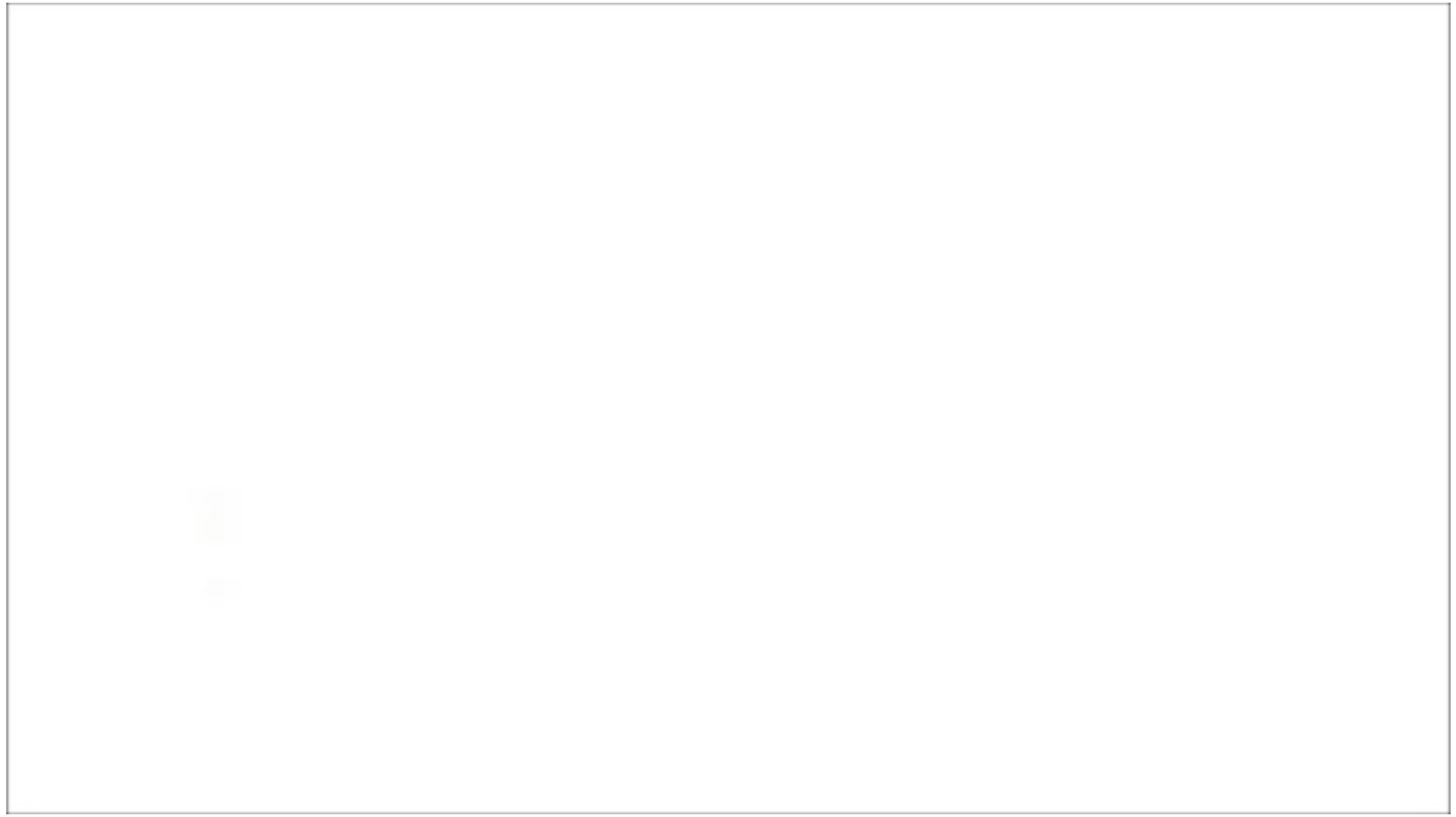
VR..

- As they walk around the virtual environment users will experience things in a similar way to the real world. For example:
- **Objects get smaller as you walk away from them** (and bigger as you move closer)
- **The direction of sounds change as you move around**
- **Objects in the virtual world appear the same dimensions as they would in the real world** (for example dogs are smaller than us but elephants are bigger

Equipment needed to create the virtual reality experience includes

Eye Goggles -	These produce the **3D images** that make up the artificial world. The goggles project slightly different views into each eye and this fools your brain into thinking that the scene is 3D. Virtual chairs look solid and so on.
Special Gloves -	The gloves **detect your hand and finger movements** which are **input into a computer** and **processed**. As users touch or use items in the virtual world, the computer can carry out these commands and make them happen. This allows the user to interact with the virtual world and perform tasks such as **moving objects** or **switching on lights** etc.
Headphones -	These control what **users hear** in the virtual world. **For example:** Distant sounds will be quieter than sounds that are close by.
Powerful Computer -	A **very powerful computer** is needed to create the virtual environment and to process/output data sent into the system by the user's actions. **For example:** The computer **produces graphics** that appear as walls, outdoor scenes and objects such as trees.

Immersion in VR



Military Training using VR



Surgery Simulation using VR




THANK
YOU
ASANTE
ASANTE
GRAZIE
NIRRINGRAZZJAK
KIA ORA
MATUR NUWUN
OBRIGADO
MULTUMESC
MOCHCHAKKERAM
OBRIGADO
SALAMAT
MAHALO
TERIMA KASIH
MATONDO
GRAZIE
KIITOS
SPASIBO
MERCI
KIITOS
MOCHCHAKKERAM
CHOKRANE
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